Hypotension / Shock

AM-05 | Adult Medical

Quick flow

01 Recognize Shock [EMT / AEMT / Paramedic]

SBP <90 mmHg or inadequate perfusion regardless of BP • confirm an abnormal automated reading manually when feasible

02 Identify the Cause [EMT / AEMT / Paramedic]

Hemorrhagic • septic • hypovolemic • cardiogenic • obstructive • anaphylactic • toxicologic • adrenal • obstetric

03 Immediate Support [EMT / AEMT / Paramedic]

Airway/ventilation • oxygen when indicated • control bleeding • prevent heat loss • position for condition

04 Monitor + Obtain Access [AEMT / Paramedic]

Cardiac monitor • glucose • repeat vitals • IV access; move rapidly to IO when critically unstable

05 Cause-Directed Resuscitation [AEMT / Paramedic]

Cause-specific reassessed fluids • hemorrhage: warmed isotonic crystalloid to perfusion when blood is unavailable • suspected TBI: adult SBP ≥110 mmHg, no permissive hypotension • treat reversible causes

06 Persistent Shock? [Paramedic]

Paramedic norepinephrine pathway • use epinephrine infusion when indicated • reassess continuously

07 Rapid Transport [EMT / AEMT / Paramedic]

Early notification • appropriate destination • consider Covenant Health Air when it provides meaningful advantage

EMT Care

Recognize shock, support airway and ventilation, control hemorrhage, prevent heat loss, and prioritize transport.

EMT

- Follow Universal Patient Care; assess mental status, pulse quality, skin signs, capillary refill, blood-pressure trend, respiratory status, and evidence of inadequate perfusion.
- Repeat an abnormal automated blood pressure manually when feasible using the correct cuff size; do not delay treatment or transport when shock is clinically evident.
- Maintain the airway, assist ventilation when needed, and give oxygen for hypoxemia, respiratory distress, shock, or another clinical indication.
- Position supine when tolerated unless respiratory distress, pulmonary edema, pregnancy, or another condition requires alternate positioning.
- Control external hemorrhage immediately using direct pressure, wound packing, tourniquets, pelvic stabilization, and other approved measures when indicated.
- Prevent heat loss, obtain frequent vital signs, identify time-critical causes, and begin rapid transport with early receiving-facility notification.

AEMT Care

All EMT care plus monitoring, vascular access, glucose assessment, and cause-appropriate crystalloid with frequent reassessment.

AEMT

- Perform all EMT actions; apply continuous cardiac monitoring, obtain blood glucose, and obtain temperature when infection, sepsis, or environmental illness is suspected.
- Establish IV access without delaying transport. When the patient is critically unstable and IV access cannot be obtained rapidly, proceed to IO access under the approved vascular-access protocol.
- For nonhemorrhagic shock without fluid-overload risk, give isotonic crystalloid 500 mL IV/IO and reassess after every aliquot.
- For heart failure, renal failure, liver failure, suspected cardiogenic shock, or another risk of volume overload, use 250 mL aliquots with reassessment after each aliquot.
- For suspected hemorrhagic shock without traumatic brain injury when blood is unavailable, titrate warmed isotonic crystalloid only to maintain a palpable radial pulse, adequate mental status, or SBP approximately 80-90 mmHg; avoid unnecessary large-volume crystalloid. Reassess after treatment.
- For suspected TBI, prevent hypotension: target adult SBP at least 110 mmHg, including age 70. Do not use permissive hypotension; use normal saline when crystalloid is required and reassess blood pressure frequently and after treatment.
- Immediately report persistent hypotension, pulmonary edema, worsening respiratory status, altered mental status, dysrhythmia, or other deterioration to the Paramedic and receiving facility.

Paramedic Care

All AEMT care plus advanced hemodynamic assessment, cause-specific intervention, vasopressor treatment, and critical destination decisions.

Paramedic

- Perform all EMT and AEMT actions; obtain and interpret a 12-lead ECG when cardiac ischemia, dysrhythmia, pulmonary embolism, or cardiogenic shock is possible.
- Use waveform capnography when ventilation is assisted and when clinically useful for trending perfusion in severe shock.
- Treat immediately reversible causes, including tension pneumothorax and unstable dysrhythmia, using the applicable linked protocol.
- For persistent hypotension or inadequate perfusion after an appropriate fluid trial - or earlier when further fluid is contraindicated - begin norepinephrine 0.05-0.5 mcg/kg/min IV/IO and titrate to adequate perfusion and MAP at least 65 mmHg.
- When norepinephrine is unavailable or significant bradycardia accompanies shock, begin epinephrine infusion 2-10 mcg/min IV/IO and titrate to adequate perfusion and MAP at least 65 mmHg.
- Use continuous ECG monitoring and frequent blood-pressure reassessment during vasopressor treatment.
- Begin transport early, notify the receiving facility, choose the closest appropriate destination for the suspected cause, and consider Covenant Health Air when it provides a meaningful time or capability advantage.

Indications

- Adult with SBP below 90 mmHg.
- Adult with suspected TBI and SBP below the 110 mmHg target; prevent hypotension rather than waiting for SBP below 90 mmHg.
- Adult with clinical evidence of inadequate perfusion regardless of blood pressure, including altered mental status, weak pulses, cool or mottled skin, delayed capillary refill, or progressive deterioration.

Contraindications

- Do not treat a single automated blood-pressure value without considering the patient's complete perfusion assessment and BP trend.

Assessment | Confirm Shock + Trend Perfusion

- Trend mental status, pulse quality, skin temperature and color, capillary refill, blood pressure or MAP, heart rate, respiratory pattern, SpO_2, and $EtCO_2$ when available.

Assessment | Identify the Cause

- Assess for external or internal hemorrhage, infection or sepsis, anaphylaxis, dehydration or gastrointestinal loss, cardiac ischemia, dysrhythmia, cardiogenic shock, pulmonary embolism, tension pneumothorax, and cardiac tamponade.
- Consider medication effect or overdose, adrenal crisis, ectopic pregnancy, and obstetric or postpartum hemorrhage.

Assessment | Monitoring

- Use continuous cardiac monitoring, pulse oximetry, serial vital signs, blood glucose, and temperature when indicated.

Treatment steps

- Establish IV access without delaying transport. In critical instability, move rapidly to IO when IV access cannot be obtained; the approved three-attempt vascular-access limitation remains applicable.
- Repeat 500 mL aliquots only while hypotension or poor perfusion persists and the patient remains fluid responsive.
- Outside the suspected-TBI branch, stop routine fluid boluses when perfusion adequately improves, SBP reaches approximately 90-100 mmHg, or MAP reaches at least 65 mmHg. The higher adult TBI SBP target of at least 110 mmHg takes precedence when TBI is suspected. In any branch, stop routine boluses for pulmonary edema, worsening respiratory status, or lack of fluid responsiveness and reassess the resuscitation strategy.
- For suspected sepsis with hypotension or inadequate perfusion, use 500 mL isotonic crystalloid aliquots and target a cumulative initial volume up to 30 mL/kg only while the patient remains fluid responsive and free of pulmonary edema. Individualize volume in patients at risk of overload and do not delay transport to complete the target volume.
- Use blood products only within an authorized, supported transfusion program with required training, monitoring, and destination coordination.
- For pregnancy beyond approximately 20 weeks, use left uterine displacement or left lateral positioning when feasible, avoid prolonged supine positioning, treat hemorrhage or another cause, and follow AO-03.
- For cardiogenic or obstructive shock, avoid repeated large-volume fluid boluses. Consider a cautious 250 mL trial only when volume depletion is possible and pulmonary edema is absent; treat tension pneumothorax or causative dysrhythmia immediately.
- For anaphylactic shock, give IM epinephrine immediately under AM-01; do not delay epinephrine for IV access or fluid administration.
- Use norepinephrine 0.05-0.5 mcg/kg/min IV/IO as the preferred vasopressor and titrate to adequate perfusion and MAP at least 65 mmHg.

Medications

Isotonic Crystalloid

500 mL IV/IO aliquots for nonhemorrhagic shock; use 250 mL aliquots when fluid-overload risk is present

- Reassess perfusion, BP or MAP, lung sounds, work of breathing, and response after every aliquot.
- For suspected sepsis, cumulative initial fluid may approach 30 mL/kg only while the patient remains fluid responsive and free of pulmonary edema.
- In hemorrhagic shock when blood is unavailable, titrate warmed isotonic crystalloid to perfusion rather than applying the nonhemorrhagic aliquot schedule; use the higher adult SBP target of at least 110 mmHg for suspected TBI. Minimize unnecessary crystalloid in uncontrolled hemorrhage and suspected cardiogenic shock.

Norepinephrine Infusion

0.05-0.5 mcg/kg/min IV/IO; titrate to adequate perfusion and MAP at least 65 mmHg

Epinephrine Infusion

2-10 mcg/min IV/IO; titrate to adequate perfusion and MAP at least 65 mmHg

Push-Dose Epinephrine

Temporary bridge only - open the dedicated Push-Dose Epinephrine reference for the approved indication, labeled 10 mcg/mL preparation, weight-based pediatric dose, and two-bolus limit

- Use the linked medication reference rather than preparing or dosing from memory.
- Begin a vasoactive infusion when indicated and contact Medical Control before any dose beyond the two-bolus limit.

Warnings / precautions

- A normal mental status or GCS of 15 does not exclude shock.
- Hypotension may be a late sign of severe shock; use the complete perfusion assessment and trend.
- The treatment goal is adequate perfusion - not normalization of blood pressure.
- Do not use indiscriminate large-volume crystalloid in uncontrolled hemorrhage, pulmonary edema, cardiogenic shock, or suspected volume overload.
- Do not use permissive hypotension when traumatic brain injury is suspected.
- Do not delay transport for repeated peripheral-IV attempts, completion of a fluid target, or normalization of blood pressure.
- Fluids and vasopressors support perfusion but do not replace correction of the underlying cause.

Clinical pearls

- A falling $EtCO_2$ in a spontaneously breathing patient may indicate worsening perfusion.
- Reassess after every fluid aliquot and vasopressor adjustment; response to treatment is part of the diagnosis.
- Early vasopressor support may be preferable to repeated fluid in cardiogenic shock, pulmonary edema, or a patient who is no longer fluid responsive.
- Control hemorrhage, decompress tension pneumothorax, treat anaphylaxis, and correct causative dysrhythmia without waiting for blood-pressure normalization.

Special populations | Suspected Traumatic Brain Injury

- Prevent hypotension: target adult SBP at least 110 mmHg, including age 70. Do not use permissive hypotension. Avoid hypoxemia and excessive crystalloid; use normal saline when crystalloid is required. Reassess blood pressure frequently and after treatment.

Special populations | Adrenal Insufficiency

- Ask about prescribed stress-dose steroids and follow the patient-specific emergency plan when authorized.

Related pathways / contacts

Push-Dose Epinephrine - Medication Reference - Open the dedicated reference for the approved indication, labeled preparation, concentration, dosing, and two-bolus limit.

UP-01 Universal Patient Care - Open universal assessment, monitoring, vascular-access, reassessment, and transport requirements.

AM-01 Allergic Reaction / Anaphylaxis - Open for anaphylactic shock and immediate IM epinephrine treatment.

AM-03 Dialysis / Renal Failure - Open for renal-failure considerations and fluid-risk guidance.

AM-07 Crashing Patient - Open for peri-arrest deterioration and resuscitation priorities.

AC-04 Acute Coronary Syndrome / STEMI - Open when myocardial ischemia or infarction may be causing shock.

AC-05 CHF / Acute Pulmonary Edema - Open for pulmonary edema or suspected acute heart failure.

AC-02 Bradycardia - Open for bradycardia causing cardiopulmonary compromise.

AC-06 Narrow-Complex Tachycardia - Open for unstable narrow-complex tachycardia.

AC-07 Monomorphic Wide-Complex Tachycardia - Open for unstable monomorphic wide-complex tachycardia.

AO-03 OB/GYN Emergency - Open for ectopic pregnancy, obstetric hemorrhage, or pregnancy-related shock.

TB-05 Head Trauma - Suspected TBI: higher adult perfusion target, serial reassessment, and controlled ventilation.

TB-12 Abdominal / Pelvic Trauma - Concealed traumatic hemorrhage and pelvic stabilization with the shared adult resuscitation strategy.

TB-06 Multiple Trauma - Open for traumatic shock, hemorrhage control, and trauma destination decisions.

TE-07 Overdose / Toxic Ingestion - Open for medication effect, overdose, or another toxicologic cause.

AR-01 Adult Airway Management - Open for advanced airway and ventilation support.

Source and references

Surviving Sepsis Campaign. International Guidelines for Management of Sepsis and Septic Shock, 2026. | American College of Surgeons Committee on Trauma, American College of Emergency Physicians, and National Association of EMS Physicians. Prehospital Hemorrhage Control and Treatment by Clinicians. | Brain Trauma Foundation. Guidelines for the Management of Severe Traumatic Brain Injury. | American Heart Association. Guidelines for CPR and Emergency Cardiovascular Care. | National Association of State EMS Officials. National Model EMS Clinical Guidelines, Version 3, 2022. | North Carolina College of Emergency Physicians. AM-05 Hypotension / Shock source protocol. | Claiborne County EMS formulary and medical-director decisions, October 2026.

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